TASK 1 SUBMISSION

Competitor Demand Prediction Using Job Postings

Understanding Data Sources and Legal

Source review, legal assessment and data collection plan for NVIDIA

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Task	Task 1 of 10 - Understanding Data Sources and Legal
Date of review	18 August 2026
Repository	github.com/Nayab-khalid/cadetX-Competitor-Demand-Prediction

Headline finding

NVIDIA's careers host permits crawling in robots.txt, but section 3.2 of NVIDIA's Terms of Service prohibits robots, scrapers, crawlers and data mining tools. Where the two disagree, the Terms of Service govern. NVIDIA's career site is therefore rejected for automated collection, and the NVIDIA data for this project is downloaded from an openly licensed public dataset instead. Checking robots.txt alone would have produced the wrong answer.

Required by Task 1	Delivered in
An approved list of data sources	Section 3 (review) and Section 4 (approved list with download links)
A clear understanding of legal requirements	Section 2
A shared plan for data collection in Task 2	Section 5

1 Where job-posting data comes from

Job-posting data reaches an analyst through five routes. They differ in legal risk, in data quality, and in whether they carry the full job description text that natural language processing needs. This survey was carried out before narrowing to the four candidate sources assessed in section 3.

Route	Examples	Strengths	Risks and limits
First-party API from the company's applicant tracking system	Greenhouse, Workday, Ashby, Lever	Authoritative, structured, real posting dates, no extraction needed	Only where the ATS exposes a public board; current vacancies only
Company career page	NVIDIA, Microsoft, Google careers	Authoritative and current	Terms of Service usually restrict automated access; no history
Open dataset platform	Kaggle, Hugging Face, GitHub	Licensed, downloadable, historical depth, often includes description text	Provenance often a third-party collection; a sample, not a census; ages quickly
Public sector open data	Cedefop Skills-OVATE, statistical offices	Openly licensed and documented	Published as aggregates, rarely company level
Commercial job-data API	Adzuna, Lightcast	Broad, cleaned, easy to query	Licensing and cost; research use often restricted

Two conclusions shaped every decision that follows. The sources most authoritative about a company are the ones most likely to restrict automated access, so authority and accessibility pull against each other. And live career pages list only currently open roles, so they carry no history: a project that must forecast hiring demand needs a historical dataset.

2 Legal and ethical requirements

Four questions decide whether a source may be used. They are answered in order, and a failure at any point rejects the source.

2.1 Terms of Service

The terms are the contractual position between the site operator and the user. They are read before any data is collected, with the date and the governing section recorded. Where terms are ambiguous the source is treated as not permitted. This rule rejected the Adzuna API, which is easy to use and has a free tier, but limits academic use to a fourteen-day trial and requires written consent for ongoing research.

2.2 robots.txt

robots.txt tells automated clients which paths they may request. It is read before collection and the applicable lines are quoted verbatim with the date, so the evidence can be re-verified. Three patterns appeared in this review: fully open, such as Anthropic; allow-listed, where everything is disallowed by default and named paths are permitted, such as Microsoft Careers; and targeted blocks, where the paths the task needs are disallowed, as with Google Careers, whose paginated results listing is blocked.

2.3 Precedence: where the two disagree

The finding that decided this task

robots.txt is an instruction to crawlers. The Terms of Service are the contractual position. They are separate instruments and they can disagree. NVIDIA is the worked example: the careers host allows the career-site path in robots.txt, while section 3.2 of NVIDIA's Terms of Service prohibits robots, spiders, scrapers, crawlers and data mining tools. Permission from the narrower instrument does not override the prohibition in the broader one. Checking robots.txt alone is not sufficient for any source, and this rule is now written into the team's legal rules for all four members.

2.4 Public versus copyrighted data

Publicly visible is not the same as free to reuse. Factual fields such as job title, posting date, location and employer attract little or no copyright protection and are analysed freely. The job description is written prose and is treated as protected expression: it is used as an input to analysis, and what this project publishes are derived features, counts, trends and forecasts, not reproductions of the source text. Licence conditions are honoured in full, including the share-alike obligation on the primary dataset.

2.5 Personal and sensitive data

No personal or sensitive data is collected, stored, processed or committed at any point. This covers names, email addresses, telephone numbers, recruiter and hiring manager identities, applicant information, and any free text identifying an individual. The controls are practical rather than declaratory:

- Any column capable of identifying a person is dropped when the file is loaded, before it is saved to disk.
- Every dataset is passed through the repository validator, which scans every cell for email addresses and telephone numbers and fails the file if it finds any.
- Endpoints connected to applications and logins are never requested, because that is where personal data appears.

3 Sources reviewed for NVIDIA

3.1 NVIDIA External Career Site - rejected

Career site: nvidia.wd5.myworkdayjobs.com/NVIDIAExternalCareerSite. Terms of Service: nvidia.com/en-us/about-nvidia/terms-of-service. Both robots.txt files and the Terms of Service were read on 18 August 2026.

```
robots.txt, nvidia.wd5.myworkdayjobs.com, read 18 August 2026

User-agent: *
Allow: /NVIDIAExternalCareerSite/
Disallow: /talentcommunity/
Disallow: /refreshFacet/
```

The main nvidia.com robots.txt allows the whole site to all crawlers. On robots.txt alone the career-site path would be permitted. The Terms of Service say otherwise. Section 3.2, Use Restrictions, prohibits any robot, spider, scraper, crawler, data mining tool or other automatic device being used to access, acquire, copy or monitor the site, and prohibits reproducing its navigational structure by automated means or an equivalent manual process. Section 3.1 permits a single copy of materials to be downloaded for personal, non-commercial internal use only. Section 4 confirms NVIDIA retains its intellectual property rights.

The site is therefore rejected for automated collection. Section 3.1 does allow a person to read the public careers page and note aggregate observations, which involves no automated tool; that is retained as a validation step only, not as a dataset. A further point is recorded for completeness: the careers site is served from a Workday host rather than from nvidia.com, so whether the nvidia.com terms formally extend to it is arguable, and the team's rule is that an arguable position is treated as not permitted.

3.2 Kaggle, LinkedIn Job Postings 2023-2024 - approved as primary

Licensed CC BY-SA 4.0, confirmed 18 August 2026. Approximately 531 MB, more than 124,000 postings across 2023 and 2024, with title, full description, location, work type, salary and application URL, plus separate company and benefits files.

This is the primary source for one decisive reason: it is the only approved source carrying the full job description text, which the NLP work in Task 3 and the skill extraction in Task 4 depend on. Its twenty-four month span also gives Task 5 and Task 7 a real time series, which no live career page can provide.

The data originates from LinkedIn, whose User Agreement prohibits automated collection. The CC BY-SA licence was applied by the uploader, not by LinkedIn. I did not collect this data and am not party to LinkedIn's agreement, but the provenance is not clean and this document says so. It is therefore labelled as third-party derived wherever it is used, filtered to NVIDIA rows on load, stripped of any identifying field, credited to Kaggle and the author, and anything derived from it is shared under the same licence.

3.3 Hugging Face, lukebarousse/data_jobs - approved as cross-check

Licensed Apache-2.0, confirmed 18 August 2026. Approximately 786,000 rows covering 2023, aggregated from several job boards, with title, company, location, posted date, salary and extracted skills.

It carries extracted skills rather than full description text, so it cannot be the input to my own preprocessing or extraction: using it as the primary source would mean inheriting somebody else's extraction and having nothing of my own to evaluate. Its value is as an independent check. If my extracted skills for NVIDIA roles diverge sharply from this dataset's, that is a signal to re-examine my method.

3.4 Adzuna API - rejected

The free tier permits 1,000 calls per month, but academic use is limited to a fourteen-day trial to validate coverage and quality, and continued research use, including aggregated vacancy counts and average salaries, requires written consent. A three-month project cannot run on a fourteen-day trial and I hold no written consent, so the source is rejected.

Also considered and not pursued: LinkedIn, Indeed and Glassdoor collected directly, because their terms prohibit it; and Cedefop Skills-OVATE, which publishes aggregated European indicators rather than company-level postings and is retained only as sector context.

4 Approved sources and download links

These are the exact sources the NVIDIA analysis will be built from, and where each one is obtained. Everything downstream in Tasks 2 to 9 comes from the two datasets below.

Source	Link	Licence	Role in the project
Kaggle: LinkedIn Job Postings 2023-2024	kaggle.com/datasets/arshkon/linkedin-job-postings	CC BY-SA 4.0	Primary. Filtered to NVIDIA rows. Feeds Tasks 2 to 9.

Source	Link	Licence	Role in the project
Hugging Face: lukebarousse/data_jobs	huggingface.co/datasets/lukebarousse/data_jobs	Apache-2.0	Cross-check on skill extraction in Task 4.
NVIDIA careers page	nvidia.com/en-us/about-nvidia/careers	Read manually under ToS 3.1	Current-state sanity check only. No automated requests, no bulk copying.

4.1 How to download the primary dataset

Open kaggle.com/datasets/arshkon/linkedin-job-postings and use the Download button, or use the Kaggle command line client, which is reproducible and preferred because the exact command can be recorded in the Data Collection Report:

```
pip install kaggle
# place your kaggle.json API token in %USERPROFILE%\.kaggle\ first
kaggle datasets download -d arshkon/linkedin-job-postings
# then unzip into work/task-2/nvidia-nayab-khalid/data/raw/
```

The archive contains a main postings file together with company and benefit lookup files. The exact file names and row counts are recorded in the Data Collection Report in Task 2, after download, rather than assumed here.

4.2 How to download the cross-check dataset

Dataset page: huggingface.co/datasets/lukebarousse/data_jobs. The single data file is data_jobs.csv, which can be downloaded directly from the page or loaded in Python:

```
import pandas as pd
url = "https://huggingface.co/datasets/lukebarousse/data_jobs/"\
      "resolve/main/data_jobs.csv"
df = pd.read_csv(url)
nvidia = df[df["company_name"].str.contains("NVIDIA", case=False, na=False)]
```

4.3 Two checks that must close before downloading

- Kaggle's Terms of Use, kaggle.com/terms, could not be retrieved on 18 August 2026. They must be read and dated before collection begins.
- The absence of personal data must be verified column by column on the downloaded files, not assumed from the field list, and then confirmed by the repository validator.

5 Data collection plan for Task 2

Step	Action	Output
1	Read and date the Kaggle Terms of Use; re-check both NVIDIA robots.txt files and the NVIDIA Terms of Service	Dated checks appended to this review
2	Download the Kaggle dataset and filter to NVIDIA rows	Working extract
3	Inspect every column for personal data and drop anything identifying	Column decision log
4	Map the retained columns onto the team schema	nvidia_postings_raw_YYYYMMDD.csv
5	Run scripts/validate_dataset.py and attach its output	Validator output
6	Cross-check volume and skills against data_jobs.csv	Comparison note

Step	Action	Output
7	Record aggregate observations from the live careers page, read manually	Current-state note

The dataset is delivered against the team schema so the four companies stay comparable. The required fields are job_id, company_name, job_title, job_description, posting_date, location and job_url, with dates in ISO-8601 format. Where the source does not supply a required field, the column is kept and left empty rather than dropped or renamed, and the gap is reported as a limitation.

The four members will not all collect from the same kind of source, because the legal position of each company's own site differs. To keep Task 6 fair, all four extract the same schema fields over the same date window, and all four also take their company's rows from the same openly licensed dataset, giving one common basis for direct comparison. The difference in provenance is stated in every cross-company comparison rather than hidden.

6 Limitations

Limitation	How it is handled
NVIDIA's own career site cannot be used for automated collection	Documented openly; an openly licensed dataset is used instead
The primary data covers 2023 and 2024, so it is historical	Every finding is stated as being about 2023 to 2024, not the present; a manual current-state observation is recorded alongside
The primary data is derived from a platform whose terms restrict collection	Labelled third-party derived throughout, used with attribution and share-alike
Coverage is a sample, not a census of NVIDIA postings	Comparisons use share of postings, never raw counts across companies
Terms and robots.txt can change during the project	All sources re-checked at the start of Task 2 and before the Task 10 submission

7 Conclusion

Four candidate sources were assessed against Terms of Service, robots.txt, copyright and personal data. NVIDIA's own career site was rejected for automated collection because its Terms of Service prohibit the tools the task would require, despite a permissive robots.txt. The Adzuna API was rejected because academic use is limited to a fourteen-day trial. The Kaggle LinkedIn Job Postings dataset, licensed CC BY-SA 4.0, is approved as the primary source because it is the only approved source carrying full description text and a genuine time series, and the Apache-2.0 licensed data_jobs dataset is approved as an independent cross-check. Both are linked in section 4 and will be downloaded once the two open checks close.

The wider lesson is now in the team's legal rules for all four members: robots.txt and the Terms of Service are separate instruments, they can disagree, and the terms govern.

Declaration	No data was collected during Task 1. No personal or sensitive data has been collected, stored or committed at any point. All checks recorded here were performed on 18 August 2026 and are repeatable.
Prepared by	Nayab Khalid, NVIDIA specialist
Date	18 August 2026

References, all accessed 18 August 2026

Source	Link
NVIDIA Terms of Service, sections 3.1, 3.2, 4	https://www.nvidia.com/en-us/about-nvidia/terms-of-service/
NVIDIA careers robots.txt	https://nvidia.wd5.myworkdayjobs.com/robots.txt
NVIDIA robots.txt	https://www.nvidia.com/robots.txt
Kaggle dataset, CC BY-SA 4.0	https://www.kaggle.com/datasets/arshkon/linkedin-job-postings
Hugging Face dataset, Apache-2.0	https://huggingface.co/datasets/lukebarousse/data_jobs
Adzuna developer terms	https://developer.adzuna.com/docs/terms_of_service
Cedefop Skills-OVATE	https://www.cedefop.europa.eu/en/tools/skills-online-vacancies

Attribution: the primary dataset is used under CC BY-SA 4.0 and credited to its author and to Kaggle, with derived work shared under the same licence. The cross-check dataset is used under Apache-2.0 with attribution.